A Residue Reformulation of the Local Curve Identity: Proof Attempt and Numerical Refutation

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Problem Statement (Conjecture 1)

Let $l_1, l_2, l_3 \in \mathbb{Z}$ satisfy

$$l_1 + l_2 + l_3 = -2, \quad l_1 \geq l_2 \geq l_3,$$

and let $\lambda_1, \lambda_2, \lambda_3$ satisfy $\lambda_1 + \lambda_2 + \lambda_3 = 0$. Set

$$X_1 = \lambda_1^{-l_1-1}\,\lambda_2^{-l_2-1}\,\lambda_3^{-l_3-1}.$$

Define

$$\bar{l} \;=\; \begin{cases} l, & l \geq 0, \\ -l-1, & l < 0, \end{cases} \qquad \sigma(m) := \tfrac{m(m+1)}{2}.$$

Then

$$Y = \frac{1}{8}\,\lambda_1^{-2l_1-1}\lambda_2^{-2l_2-1}\lambda_3^{-2l_3-1}\Big(\sigma(\bar{l}_1)\lambda_1^{-2}+\sigma(\bar{l}_2)\lambda_2^{-2}+\sigma(\bar{l}_3)\lambda_3^{-2}+l_1l_2\,\lambda_1^{-1}\lambda_2^{-1}+l_2l_3\,\lambda_2^{-1}\lambda_3^{-1}+l_1l_3\,\lambda_1^{-1}\lambda_3^{-1}\Big).$$

Further, define

$$X_2 = -\lambda_1^{-2l_1-2}\lambda_2^{-2l_2-2}(\lambda_1+\lambda_2)^{-2l_3-2}\left(\sum_{\substack{1\leq k\leq l_1 \\ k\equiv l_1 \pmod 2}} A(l_1,l_2,l_3,k)+\sum_{\substack{1\leq k\leq l_2 \\ k\equiv l_2 \pmod 2}} B(l_1,l_2,l_3,k)\right),$$

where

$$\begin{aligned} A(l_1,l_2,l_3,k) = \operatorname{Res}_{h=0}\Big[&h^{-k}(h-\lambda_1)^2(h+\lambda_2)^{k+l_2}(h+\lambda_1-\lambda_2)^{k+l_3} \\ &\times (h-\lambda_1+\lambda_2)^{l_1-l_2-k}(h-2\lambda_1-\lambda_2)^{l_1-l_3-k}(h-2\lambda_1)^{k-2-2l_1}\Big], \end{aligned}$$

and

$$\begin{aligned} B(l_1,l_2,l_3,k) = \operatorname{Res}_{h=0}\Big[&h^{-k}(h-\lambda_2)^2(h+\lambda_1)^{k+l_1}(h+\lambda_2-\lambda_1)^{k+l_3} \\ &\times (h-\lambda_2+\lambda_1)^{l_2-l_1-k}(h-2\lambda_2-\lambda_1)^{l_2-l_3-k}(h-2\lambda_2)^{k-2-2l_2}\Big]. \end{aligned}$$

Claim: $Y = X_2 + \frac{1}{8}\,X_1$.

Section 1: Normalization (Equivalence Reduction)

Let

$$X_1 = \lambda_1^{-l_1-1}\lambda_2^{-l_2-1}\lambda_3^{-l_3-1}, \qquad X_2 = -\lambda_1^{-2l_1-2}\lambda_2^{-2l_2-2}(\lambda_1+\lambda_2)^{-2l_3-2}\, R,$$

$$Y = \frac{1}{8}\,\lambda_1^{-2l_1-1}\lambda_2^{-2l_2-1}\lambda_3^{-2l_3-1}\, Q.$$

Assume for the moment that the identity

$$Y \;=\; X_2 + \frac{1}{8}\,X_1 \tag{1}$$

holds. We shall extract from it the asserted formula

$$R = \frac{1}{8}\Big(\lambda_1^{l_1+1}\lambda_2^{l_2+1}\lambda_3^{l_3+1} - \lambda_1\lambda_2\lambda_3\, Q\Big). \tag{2}$$

(The converse implication will then be immediate by reversing the steps.)

———————————————————— 1. Pull out a common monomial factor ————————————————————

Set

$$M := \lambda_1^{-2l_1-2}\lambda_2^{-2l_2-2}\lambda_3^{-2l_3-2}.$$

Because $-2l_3-2$ is even and $\lambda_3 = -\lambda_1-\lambda_2$, we have

$$(\lambda_1+\lambda_2)^{-2l_3-2} = (-\lambda_3)^{-2l_3-2} = \lambda_3^{-2l_3-2}. \tag{3}$$

Write each term of (1) with the factor M:

- For Y:

$$Y = \frac{1}{8}\,\lambda_1^{-2l_1-1}\lambda_2^{-2l_2-1}\lambda_3^{-2l_3-1}Q = \frac{1}{8}\,M\,(\lambda_1\lambda_2\lambda_3)\,Q. \tag{4}$$

- For $\frac{1}{8}X_1$:

$$\frac{1}{8}X_1 = \frac{1}{8}\,\lambda_1^{-l_1-1}\lambda_2^{-l_2-1}\lambda_3^{-l_3-1} = \frac{1}{8}\,M\;\lambda_1^{l_1+1}\lambda_2^{l_2+1}\lambda_3^{l_3+1}. \tag{5}$$

- For X_2: apply (3):

$$X_2 = -\lambda_1^{-2l_1-2}\lambda_2^{-2l_2-2}\lambda_3^{-2l_3-2}R = -M\,R. \tag{6}$$

———————————————————— 2. Substitute in (1) ————————————————————

Insert (4)–(6) into $Y = X_2 + \frac{1}{8}X_1$:

$$\frac{1}{8}\,M(\lambda_1\lambda_2\lambda_3 Q) \;=\; -M\,R \;+\; \frac{1}{8}\,M\,\lambda_1^{l_1+1}\lambda_2^{l_2+1}\lambda_3^{l_3+1}.$$

Divide by the non-zero common factor M:

$$\frac{1}{8}\big(\lambda_1\lambda_2\lambda_3 Q\big) \;=\; -R \;+\; \frac{1}{8}\,\lambda_1^{l_1+1}\lambda_2^{l_2+1}\lambda_3^{l_3+1}.$$

3. Isolate R

Re-arranging gives

$$R = \frac{1}{8}\Big(\lambda_1^{l_1+1}\lambda_2^{l_2+1}\lambda_3^{l_3+1} - \lambda_1\lambda_2\lambda_3\, Q\Big),$$

which is exactly (2).

4. Converse implication

Conversely, if (2) holds, start from (6) and substitute the right-hand side of (2); then reverse the algebraic steps above to obtain (1). Therefore (1) and (2) are equivalent, completing the proof. $\square$

Section 2: Boundary Cases ($l_1 \leq 0$)

Solution.

Assume throughout that the negative powers appearing in Q are meaningful, i.e. that $\lambda_1, \lambda_2, \lambda_3 \neq 0$. Since $\lambda_3 = -\lambda_1 - \lambda_2$, this means in particular that $\lambda_1, \lambda_2, \lambda_1 + \lambda_2$ are nonzero.

We are given

$$l_1 + l_2 + l_3 = -2, \qquad l_1 \geq l_2 \geq l_3, \qquad l_1 \leq 0.$$

Because $l_1 \geq l_2 \geq l_3$, we have

$$l_2 \leq l_1 \leq 0, \qquad l_3 \leq l_2 \leq 0.$$

Thus all three integers l_1, l_2, l_3 are nonpositive. If $l_1 \leq -1$, then

$$l_1 \leq -1, \qquad l_2 \leq -1, \qquad l_3 \leq -1,$$

so

$$l_1 + l_2 + l_3 \leq -3,$$

contradicting $l_1 + l_2 + l_3 = -2$. Therefore

$$l_1 = 0.$$

Then

$$l_2 + l_3 = -2,$$

with

$$0 = l_1 \geq l_2 \geq l_3.$$

Since $l_3 = -2 - l_2$, the condition $l_2 \geq l_3$ gives

$$l_2 \geq -2 - l_2,$$

hence

$$2l_2 \geq -2,$$

so

$$l_2 \geq -1.$$

Also $l_2 \leq 0$. Since $l_2 \in \mathbb{Z}$, the only possibilities are

$$l_2 = 0 \quad \text{or} \quad l_2 = -1.$$

If $l_2 = 0$, then $l_3 = -2$, giving

$$(l_1, l_2, l_3) = (0, 0, -2).$$

If $l_2 = -1$, then $l_3 = -1$, giving

$$(l_1, l_2, l_3) = (0, -1, -1).$$

Thus the only possible triples are

$$(l_1, l_2, l_3) \in \{(0, 0, -2), (0, -1, -1)\}.$$

Now we verify the two cases.
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Case 1: $(l_1, l_2, l_3) = (0, 0, -2)$
The parity-restricted sums are

$$\sum_{\substack{1 \leq k \leq l_1 \\ k \equiv l_1 \pmod 2}} A(\ldots, k) = \sum_{\substack{1 \leq k \leq 0 \\ k \equiv 0 \pmod 2}} A(\ldots, k),$$

and

$$\sum_{\substack{1 \leq k \leq l_2 \\ k \equiv l_2 \pmod 2}} B(\ldots, k) = \sum_{\substack{1 \leq k \leq 0 \\ k \equiv 0 \pmod 2}} B(\ldots, k).$$

Both indexing sets are empty, so both sums are equal to 0.
Next compute Q. We have

$$\bar{l}_1 = \bar{0} = 0, \qquad \bar{l}_2 = \bar{0} = 0,$$

and since $l_3 = -2 < 0$,

$$\bar{l}_3 = -l_3 - 1 = 2 - 1 = 1.$$

Also

$$\sigma(0) = 0, \qquad \sigma(1) = \frac{1 \cdot 2}{2} = 1.$$

Moreover,

$$l_1 l_2 = 0 \cdot 0 = 0, \qquad l_2 l_3 = 0 \cdot (-2) = 0, \qquad l_1 l_3 = 0 \cdot (-2) = 0.$$

Therefore

$$Q = 0 \cdot \lambda_1^{-2} + 0 \cdot \lambda_2^{-2} + 1 \cdot \lambda_3^{-2} + 0 + 0 + 0,$$

so

$$Q = \lambda_3^{-2}.$$

Hence

$$\lambda_1 \lambda_2 \lambda_3 Q = \lambda_1 \lambda_2 \lambda_3 \cdot \lambda_3^{-2} = \lambda_1 \lambda_2 \lambda_3^{-1}.$$

On the other hand,

$$\lambda_1^{l_1+1} \lambda_2^{l_2+1} \lambda_3^{l_3+1} = \lambda_1^{0+1} \lambda_2^{0+1} \lambda_3^{-2+1} = \lambda_1 \lambda_2 \lambda_3^{-1}.$$

Thus

$$\lambda_1^{l_1+1} \lambda_2^{l_2+1} \lambda_3^{l_3+1} = \lambda_1 \lambda_2 \lambda_3 Q.$$

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Case 2: $(l_1, l_2, l_3) = (0, -1, -1)$
The first parity-restricted sum is

$$\sum_{\substack{1 \le k \le l_1 \\ k \equiv l_1 \pmod 2}} A(\dots, k) = \sum_{\substack{1 \le k \le 0 \\ k \equiv 0 \pmod 2}} A(\dots, k),$$

which is empty. The second is

$$\sum_{\substack{1 \le k \le l_2 \\ k \equiv l_2 \pmod 2}} B(\dots, k) = \sum_{\substack{1 \le k \le -1 \\ k \equiv -1 \pmod 2}} B(\dots, k),$$

which is also empty. Hence both sums are 0.
Now compute Q. We have

$$\bar{l}_1 = \bar{0} = 0.$$

Since $l_2 = l_3 = -1 < 0$,

$$\bar{l}_2 = -l_2 - 1 = 1 - 1 = 0,$$

and

$$\bar{l}_3 = -l_3 - 1 = 1 - 1 = 0.$$

Therefore

$$\sigma(\bar{l}_1) = \sigma(\bar{l}_2) = \sigma(\bar{l}_3) = \sigma(0) = 0.$$

The pairwise products are

$$l_1 l_2 = 0 \cdot (-1) = 0,$$

$$l_2 l_3 = (-1)(-1) = 1,$$

and

$$l_1 l_3 = 0 \cdot (-1) = 0.$$

Thus

$$Q = 0 + 0 + 0 + 0 + 1 \cdot \lambda_2^{-1} \lambda_3^{-1} + 0,$$

so

$$Q = \lambda_2^{-1} \lambda_3^{-1}.$$

Hence

$$\lambda_1 \lambda_2 \lambda_3 Q = \lambda_1 \lambda_2 \lambda_3 \cdot \lambda_2^{-1} \lambda_3^{-1} = \lambda_1.$$

On the other hand,

$$\lambda_1^{l_1+1} \lambda_2^{l_2+1} \lambda_3^{l_3+1} = \lambda_1^{0+1} \lambda_2^{-1+1} \lambda_3^{-1+1} = \lambda_1 \lambda_2^0 \lambda_3^0 = \lambda_1.$$

Therefore

$$\lambda_1^{l_1+1} \lambda_2^{l_2+1} \lambda_3^{l_3+1} = \lambda_1 \lambda_2 \lambda_3 Q.$$

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Combining the enumeration with the two direct verifications, we conclude that if

$$l_1 + l_2 + l_3 = -2, \qquad l_1 \geq l_2 \geq l_3, \qquad l_1 \leq 0,$$

then necessarily

$$(l_1, l_2, l_3) \in \{(0, 0, -2), (0, -1, -1)\},$$

both parity-restricted sums defining R are empty, so

$$R = 0,$$

and in both cases

$$\lambda_1^{l_1+1}\lambda_2^{l_2+1}\lambda_3^{l_3+1} = \lambda_1\lambda_2\lambda_3 Q.$$

This proves the statement.

Section 3: Compressing the Residue Sums

We prove both identities by a direct factorization and finite linearity of coefficient extraction.

Work formally in a Laurent-series setting, for instance over the coefficient field $K = \mathbb{Q}(\lambda_1, \lambda_2)$. The residue map

$$\mathrm{Res}_{h=0} : K((h)) \to K, \qquad \mathrm{Res}_{h=0}[f(h)] = [h^{-1}]f(h),$$

is K-linear. In particular, for any finite set S and any Laurent series $f_k(h)$,

$$\sum_{k\in S} \mathrm{Res}_{h=0}[f_k(h)] = \mathrm{Res}_{h=0}\left[\sum_{k\in S} f_k(h)\right].$$

The finiteness of S is the only point needed here; no convergence issue is involved.

The A-sum

Let

$$S_A = \{k \in \mathbb{Z} : 1 \le k \le l_1,\ k \equiv l_1 \pmod 2\}.$$

For $k \in S_A$, compute $P_A(h)q_A(h)^k$. By definition,

$$q_A(h)^k = \frac{(h+\lambda_2)^k(h+\lambda_1-\lambda_2)^k(h-2\lambda_1)^k}{h^k(h-\lambda_1+\lambda_2)^k(h-2\lambda_1-\lambda_2)^k}.$$

Therefore

$$\begin{aligned} P_A(h)q_A(h)^k = {} & (h-\lambda_1)^2(h+\lambda_2)^{l_2}(h+\lambda_1-\lambda_2)^{l_3}(h-\lambda_1+\lambda_2)^{l_1-l_2} \\ & \times (h-2\lambda_1-\lambda_2)^{l_1-l_3}(h-2\lambda_1)^{-2-2l_1} \\ & \times \frac{(h+\lambda_2)^k(h+\lambda_1-\lambda_2)^k(h-2\lambda_1)^k}{h^k(h-\lambda_1+\lambda_2)^k(h-2\lambda_1-\lambda_2)^k}. \end{aligned}$$

Combining like bases using the usual exponent laws for integer powers gives

$$\begin{aligned} P_A(h)q_A(h)^k = {} & h^{-k}(h-\lambda_1)^2(h+\lambda_2)^{k+l_2}(h+\lambda_1-\lambda_2)^{k+l_3} \\ & \times (h-\lambda_1+\lambda_2)^{l_1-l_2-k}(h-2\lambda_1-\lambda_2)^{l_1-l_3-k}(h-2\lambda_1)^{k-2-2l_1}. \end{aligned}$$

The right-hand side is exactly the integrand appearing in the definition of $A(l_1, l_2, l_3, k)$. Hence

$$A(l_1, l_2, l_3, k) = \mathrm{Res}_{h=0}\big[P_A(h) q_A(h)^k\big] .$$

Now sum over the parity-restricted finite set S_A. By linearity of $\mathrm{Res}_{h=0}$,

$$\begin{aligned}\sum_{\substack{1 \le k \le l_1 \\ k \equiv l_1 \pmod 2}} A(l_1, l_2, l_3, k) &= \sum_{k \in S_A} \mathrm{Res}_{h=0}\big[P_A(h) q_A(h)^k\big] \\ &= \mathrm{Res}_{h=0}\left[\sum_{k \in S_A} P_A(h) q_A(h)^k\right] \\ &= \mathrm{Res}_{h=0}\left[P_A(h) \sum_{k \in S_A} q_A(h)^k\right] .\end{aligned}$$

But by definition,

$$\Phi_{l_1}(z) = \sum_{\substack{1 \le k \le l_1 \\ k \equiv l_1 \pmod 2}} z^k,$$

so substituting $z = q_A(h)$ gives

$$\sum_{k \in S_A} q_A(h)^k = \Phi_{l_1}(q_A(h)).$$

Therefore

$$\boxed{\sum_{\substack{1 \le k \le l_1 \\ k \equiv l_1 \pmod 2}} A(l_1, l_2, l_3, k) = \mathrm{Res}_{h=0}[P_A(h)\, \Phi_{l_1}(q_A(h))] .}$$

The B-sum

Assume now that $l_2 \ge 1$, and define

$$S_B = \{k \in \mathbb{Z} : 1 \le k \le l_2,\ k \equiv l_2 \pmod 2\}.$$

For $k \in S_B$, we similarly compute

$$q_B(h)^k = \frac{(h + \lambda_1)^k (h + \lambda_2 - \lambda_1)^k (h - 2\lambda_2)^k}{h^k (h - \lambda_2 + \lambda_1)^k (h - 2\lambda_2 - \lambda_1)^k}.$$

Thus

$$\begin{aligned}P_B(h) q_B(h)^k = {} & (h - \lambda_2)^2 (h + \lambda_1)^{l_1} (h + \lambda_2 - \lambda_1)^{l_3} (h - \lambda_2 + \lambda_1)^{l_2 - l_1} \\ & \times (h - 2\lambda_2 - \lambda_1)^{l_2 - l_3} (h - 2\lambda_2)^{-2 - 2l_2} \\ & \times \frac{(h + \lambda_1)^k (h + \lambda_2 - \lambda_1)^k (h - 2\lambda_2)^k}{h^k (h - \lambda_2 + \lambda_1)^k (h - 2\lambda_2 - \lambda_1)^k}.\end{aligned}$$

Combining powers gives

$$\begin{aligned}P_B(h)q_B(h)^k &= h^{-k}(h-\lambda_2)^2(h+\lambda_1)^{k+l_1}(h+\lambda_2-\lambda_1)^{k+l_3}\\ &\quad\times(h-\lambda_2+\lambda_1)^{l_2-l_1-k}(h-2\lambda_2-\lambda_1)^{l_2-l_3-k}(h-2\lambda_2)^{k-2-2l_2}.\end{aligned}$$

This is exactly the integrand defining $B(l_1,l_2,l_3,k)$. Hence

$$B(l_1,l_2,l_3,k)=\mathrm{Res}_{h=0}\big[P_B(h)q_B(h)^k\big].$$

Using finite linearity of the residue,

$$\begin{aligned}\sum_{\substack{1\le k\le l_2\\ k\equiv l_2 \pmod 2}} B(l_1,l_2,l_3,k) &= \sum_{k\in S_B}\mathrm{Res}_{h=0}\big[P_B(h)q_B(h)^k\big]\\ &= \mathrm{Res}_{h=0}\left[\sum_{k\in S_B}P_B(h)q_B(h)^k\right]\\ &= \mathrm{Res}_{h=0}\left[P_B(h)\sum_{k\in S_B}q_B(h)^k\right]\\ &= \mathrm{Res}_{h=0}[P_B(h)\Phi_{l_2}(q_B(h))].\end{aligned}$$

Therefore

$$\boxed{\sum_{\substack{1\le k\le l_2\\ k\equiv l_2 \pmod 2}} B(l_1,l_2,l_3,k)=\mathrm{Res}_{h=0}[P_B(h)\,\Phi_{l_2}(q_B(h))].}$$

Both desired compressed residue identities follow.

Section 4: Numerical Verification and Counterexample

Status: The identity $Y=X_2+\frac{1}{8}X_1$ as stated in Conjecture 1 is **false** for $l_1\ge 1$ under the given formulas for A and B. Sections 1–3 establish correct structural lemmas; the central claim (Section 4) fails numerically.

Explicit counterexample. Take

$$l_1=1,\quad l_2=0,\quad l_3=-3,\qquad \lambda_1=1,\quad \lambda_2=2,\quad \lambda_3=-3.$$

One checks $l_1+l_2+l_3=-2$, $l_1\ge l_2\ge l_3$, and $\lambda_1+\lambda_2+\lambda_3=0$.

Computation of $A(1,0,-3,1)$. The sum defining R has a single term $k=1$ (since $1\le k\le l_1=1$ and $k\equiv l_1=1 \pmod 2$), and $\sum B=0$ since $l_2=0$.

Substituting into the formula for A:

$$\begin{aligned}A(1,0,-3,1)&=\mathrm{Res}_{h=0}\big[h^{-1}(h-1)^2(h+2)^1(h-1)^{-2}(h+1)^0(h-4)^3(h-2)^{-3}\big]\\ &=\mathrm{Res}_{h=0}\big[h^{-1}(h+2)(h-4)^3(h-2)^{-3}\big].\end{aligned}$$

Since the remaining factors $(h+2)(h-4)^3(h-2)^{-3}$ evaluate to

$$(0+2)(0-4)^3(0-2)^{-3} = 2 \cdot (-64) \cdot (-\tfrac{1}{8}) = 16$$

at $h = 0$, we get $A(1, 0, -3, 1) = 16$.

Computation of X_2.

$$X_2 = -\lambda_1^{-4}\lambda_2^{-2}(\lambda_1 + \lambda_2)^4 \cdot 16 = -(1) \cdot \tfrac{1}{4} \cdot 81 \cdot 16 = -324.$$

Computation of X_1 and $(1/8)X_1$.

$$X_1 = \lambda_1^{-2}\lambda_2^{-1}\lambda_3^2 = 1 \cdot \tfrac{1}{2} \cdot 9 = \tfrac{9}{2}, \qquad \tfrac{1}{8}X_1 = \tfrac{9}{16}.$$

Right-hand side.

$$X_2 + \tfrac{1}{8}X_1 = -324 + \tfrac{9}{16} = -\frac{5175}{16}.$$

Computation of Y. We have $\bar{l}_1 = 1$, $\sigma(1) = 1$; $\bar{l}_2 = 0$, $\sigma(0) = 0$; $\bar{l}_3 = 2$, $\sigma(2) = 3$; and $l_1 l_3 = -3$. Hence

$$Q = 1 \cdot \lambda_1^{-2} + 3 \cdot \lambda_3^{-2} + (-3) \cdot \lambda_1^{-1}\lambda_3^{-1} = 1 + \tfrac{1}{3} + 1 = \tfrac{7}{3}.$$

Therefore

$$Y = \tfrac{1}{8}\,\lambda_1^{-3}\lambda_2^{-1}\lambda_3^5\,Q = \tfrac{1}{8} \cdot 1 \cdot \tfrac{1}{2} \cdot (-243) \cdot \tfrac{7}{3} = -\frac{567}{16}.$$

Conclusion.

$$Y = -\frac{567}{16} \;\neq\; -\frac{5175}{16} = X_2 + \tfrac{1}{8}X_1.$$

The identity fails by $\frac{567-5175}{16} = -288$.

Systematic failure. Exact arithmetic (using Python's `fractions` module) confirms the identity also fails for: $(l_1, l_2, l_3) \in \{(1, -1, -2), (2, 0, -4), (2, 1, -5)\}$ with $(\lambda_1, \lambda_2) = (1, 2)$; and for $(1, 0, -3)$ with $(\lambda_1, \lambda_2) = (1, 3)$. The boundary cases $l_1 = 0$ (i.e., $(0, 0, -2)$ and $(0, -1, -1)$) are verified correct by Section 2, as both residue sums are empty there.

Diagnosis. The formulas for A and B and the prefactor $(\lambda_1 + \lambda_2)^{-2l_3-2}$ in X_2, as transcribed from the OCR'd problem statement, appear to contain a typographic error in the exponent of $(\lambda_1 + \lambda_2)$. Numerical evidence suggests the exponent $-2l_3 - 2$ should instead be $-l_3 - 1 = l_1 + l_2 + 1$ in the cases where $l_2 \leq 0$, but even this substitution fails for $l_2 \geq 1$, indicating a deeper error in the formulas for A and/or B themselves.

The three structural lemmas — normalization (Section 1), boundary-case verification (Section 2), and compression of parity-restricted sums into single residues (Section 3) — are all correct and verified, and would remain valid for any corrected version of the conjecture.